

Chapter 7: Energy of a system

Systems and environments

Kinetic energy

Potential energy

Conservative and non-conservative forces Energy Diagrams

Work done by a force

Work KE Theorem

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Introduction to Energy

- ▶ A variety of problems can be solved with Newton's Laws and associated principles.
- ▶ Some problems that could theoretically be solved with Newton's Laws are very difficult in practice.
 - These problems can be made easier with other techniques.
- ▶ The concept of energy is one of the most important topics in science and engineering.
- ▶ Every physical process that occurs in the Universe involves **energy and energy transfers or transformations.**
- ▶ **Energy is not easily defined.**

Introduction

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Analysis Model

- ▶ The new approach will involve changing from a **particle model to a system model**.
- ▶ In this chapter, systems are introduced along with three ways to store energy in a system.

Introduction

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Systems

- ▶ A **system** is a small portion of the Universe.
 - We will ignore the details of the rest of the Universe.
- ▶ A critical skill is to **identify** the system.
 - The first step to take in solving a problem
- ▶ A **valid** system:
 - May be a single object or particle
 - May be a collection of objects or particles
 - May be a region of space
 - May vary with time in size and shape

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Problem Solving Notes

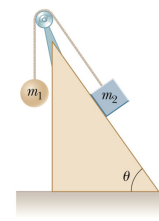
- ▶ The general problem solving approach may be used with an addition to the categorize step.
- ▶ Categorize step of general strategy
 - Identify the need for a system approach
 - Identify the particular system
 - Also identify a system boundary
 - An imaginary surface that divides the Universe into the system and the environment
 - Not necessarily coinciding with a real surface
 - The environment surrounds the system

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System Example

- ▶ A force applied to an object in empty space
 - System is the object
 - Its surface is the system boundary
 - The force is an influence on the system from its environment that acts across the system boundary.
- ▶ Ball and mass on incline
 - **System** ball, block, cord.
 - **Influence** – gravitational, normal, and frictional forces, force exerted by pulley on cord.
 - Force of cord on ball and block (tension) are **internal forces** – not an influence.

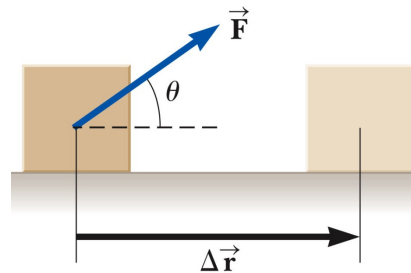


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Work

- ▶ The **work**, W , done **on** a system **by** an agent exerting a **constant force** on the system



- ▶ $W = F \Delta r \cos \theta$
 - The displacement is that of the point of application of the force.
 - A force does no work on the object if the force does not move through a displacement.
 - The work done by a force on a moving object is zero when the force applied is perpendicular to the displacement of its point of application.

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Displacement in the Work Equation

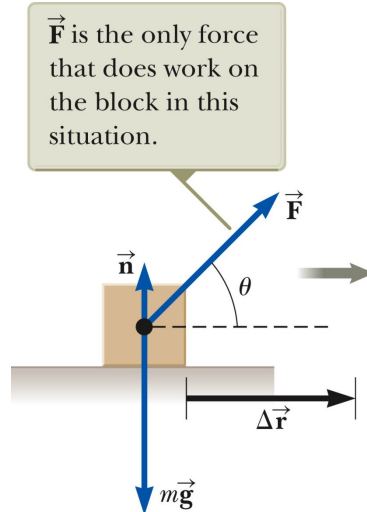
- ▶ If the force is applied to a **rigid object** that can be modeled as a particle, the displacement is the same as that of the particle.
- ▶ For a **deformable system**, the displacement of the object generally is not the same as the displacement associated with the forces applied (e.g. Balloon).

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Work Example

- ▶ The normal force and the gravitational force do no work on the object.
 - $\cos \theta = \cos 90^\circ = 0$
- ▶ The force $\vec{F}$ is the only force that does work on the object.
- ▶ Hold an object at arms length for 3 minutes?



More About Work

- ▶ The **sign** of the work depends on the direction of the force relative to the displacement.
 - Work is **positive** when projection of $\vec{F}$ onto $\Delta\vec{r}$ is in the same direction as the displacement.
 - Work is **negative** when the projection is in the opposite direction.
- ▶ The work done by a force can be calculated, but that **force is not necessarily the cause of the displacement.**
- ▶ Work is a **scalar quantity**.
- ▶ The unit of work is a **joule (J)**
 - $1 \text{ joule} = 1 \text{ newton} \cdot 1 \text{ meter} = \text{kg} \cdot \text{m}^2/\text{s}^2$
 - $\text{J} = \text{N} \cdot \text{m}$

Work Is An Energy Transfer

- ▶ If the work done on a system is positive – energy is **transferred to** the system.
- ▶ If the work done on the system is negative – energy is **transferred from** the system.
- ▶ If a system interacts with its environment, this interaction can be described as a **transfer of energy across the system boundary**.
 - This will result in a change in the amount of energy stored in the system.

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Scalar Product of Two Vectors

- ▶ The **scalar product** of two vectors is written as

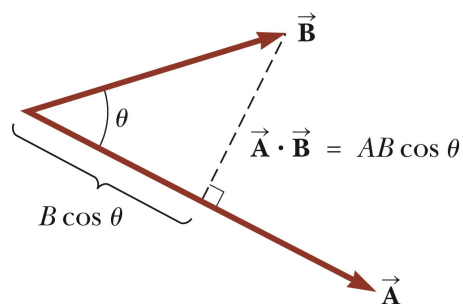
$$\vec{A} \cdot \vec{B} \equiv AB \cos \theta$$

- ▶ Applied to work, this means

$$W = F \Delta r \cos \theta = \vec{F} \cdot \Delta \vec{r}$$

- ▶ Can also use the components:

$$W = F_x r_x + F_y r_y + F_z r_z$$



Example 7.3

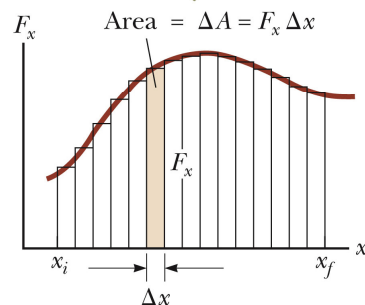
- ▶ A particle moving in the xy plane undergoes a displacement given by $\Delta\vec{r} = (2.0\hat{i} + 3.0\hat{j})$ m as a constant force $\vec{F} = (5.0\hat{i} + 2.0\hat{j})$ N acts on the particle. Calculate the work done by the force on the particle.

Work Done by a Varying Force

- ▶ To use $W = F\Delta r \cos\theta$, the force must be constant, so the equation cannot be used to calculate the work done by a varying force.
- ▶ Assume that during a very small displacement, Δx , F is constant.
- ▶ For that displacement, $W \sim F\Delta x$
- ▶ For all of the intervals,

$$W \approx \sum_{x_i}^{x_f} F_x \Delta x$$

The total work done for the displacement from x_i to x_f is approximately equal to the sum of the areas of all the rectangles.



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Work Done by a Varying Force, cont.

- As the small displacements approach zero.

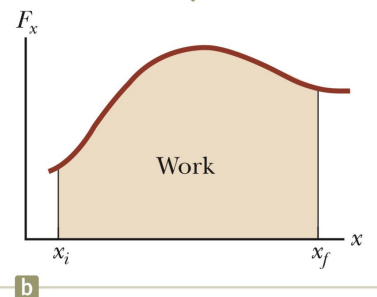
$$\lim_{\Delta x \rightarrow 0} \sum_{x_i}^{x_f} F_x \Delta x = \int_{x_i}^{x_f} F_x dx$$

- Therefore,

$$W = \int_{x_i}^{x_f} F_x dx$$

- The work done is equal to the area under the curve between x_i and x_f .

The work done by the component F_x of the varying force as the particle moves from x_i to x_f is *exactly* equal to the area under the curve.



Work Done By Multiple Forces

- If **more than one force acts on a system** and the system can be modeled as a particle, the total work done on the system is the **work done by the net force**.

$$\sum W = W_{\text{ext}} = \int_{x_i}^{x_f} (\sum F_x) dx$$

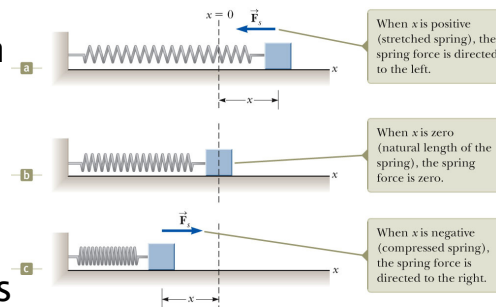
- If the **system cannot be modeled as a particle**, then the total work is equal to the algebraic sum of the work done by the individual forces.

$$\sum W = W_{\text{ext}} = \sum_{\text{forces}} \left(\int \vec{F} \cdot d\vec{r} \right)$$

- Remember work is a scalar, so this is the **algebraic sum**.

Work Done By A Spring

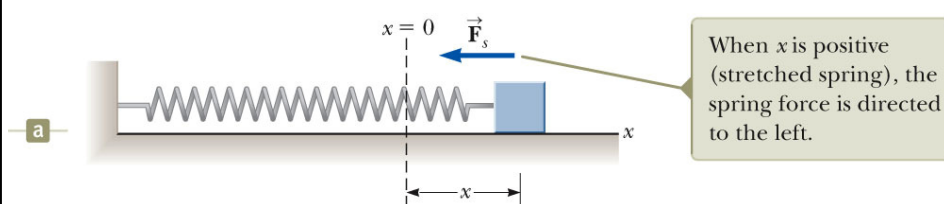
- ▶ A model of a common physical system for which the **force varies** with position is a spring (and block).
- ▶ The **block** is on a horizontal, frictionless surface.



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Spring Force (Hooke's Law)



- ▶ The force exerted by the spring is

$$F_s = -kx$$

- x is the **position** of the block with respect to the equilibrium position ($x = 0$).
- k is called the **spring constant** and measures the stiffness of the spring.
- ▶ This is called **Hooke's Law**.

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Hooke's Law cont.

- ▶ The force exerted by the spring is always directed **opposite to the displacement** from equilibrium.
- ▶ The spring force is sometimes called the **restoring force**.
- ▶ If the block is released it will **oscillate** back and forth between $-x$ and x .

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Work Done by a Spring

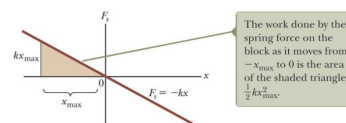
- ▶ Identify the **block as the system**.
- ▶ Calculate the work by spring as the block moves from $x_i = -x_{\max}$

$x_f = 0$.

$$W_s = \int \vec{F}_s \cdot d\vec{r} = \int_{x_i}^{x_f} (-kx\hat{i}) \cdot (dx\hat{i})$$

$$= \int_{-x_{\max}}^0 (-kx) dx = \frac{1}{2} kx_{\max}^2$$

- ▶ Calculate the work as the block moves from $x_i = 0$ to $x_f = x_{\max}$. $W = -\frac{1}{2} kx^2$
- ▶ The net work done as the block moves from $-x_{\max}$ to $x_{\max}$ is zero.
- ▶ Area under the curve.



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Work Done by a Spring cont.

- Assume the block undergoes an arbitrary displacement from $x = x_i$ to $x = x_f$.
- The **work done by the spring on the block** is

$$W_s = \int_{x_i}^{x_f} (-kx) dx = \frac{1}{2}kx_i^2 - \frac{1}{2}kx_f^2$$

- If the motion ends where it begins, $W = 0$ J

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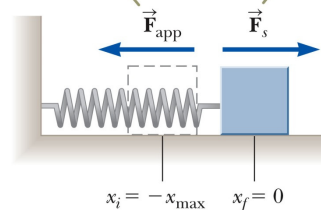
Spring with an Applied Force

- Suppose an external agent, F_{app} , stretches the spring (controlled).
- The applied force is **equal and opposite** to the spring force.

$$\vec{F}_{app} = F_{app} \hat{i} = -\vec{F}_s = -(-kx\hat{i}) = kx\hat{i}$$
- Work done by F_{app} as the block moves from $-x_{max}$ to $x = 0$ is equal to $\frac{1}{2} kx_{max}^2$
- For any displacement, the work done by the applied force is

$$W_{app} = \int_{x_i}^{x_f} (kx) dx = \frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2$$

If the process of moving the block is carried out very slowly, then $\vec{F}_{app}$ is equal in magnitude and opposite in direction to $\vec{F}_s$ at all times.



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Kinetic Energy

- ▶ A possible result of work acting as an influence on a system is that the **system changes its speed**.
- ▶ The system could possess **kinetic energy**.
- ▶ Kinetic Energy is the energy of a particle due to its motion.
 - $K = \frac{1}{2} mv^2$
 - K is the kinetic energy
 - m is the mass of the particle
 - v is the speed of the particle
- ▶ **A change in kinetic energy is one possible result of doing work to transfer energy into a system.**

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Kinetic Energy, cont

- ▶ Calculating the work:

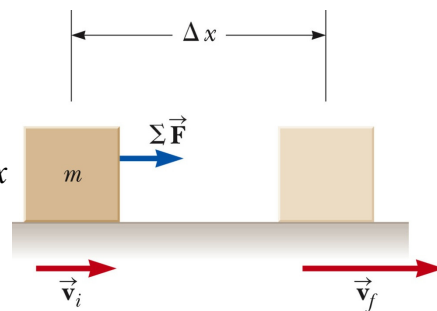
$$W_{ext} = \int_{x_i}^{x_f} \sum F \, dx = \int_{x_i}^{x_f} ma \, dx$$

$$W_{ext} = \int_{x_i}^{x_f} m \frac{dv}{dt} dx = \int_{x_i}^{x_f} m \frac{dv}{dx} \frac{dx}{dt} dx$$

$$W_{ext} = \int_{v_i}^{v_f} mv \, dv$$

$$W_{ext} = \frac{1}{2} mv_f^2 - \frac{1}{2} mv_i^2$$

$$W_{ext} = K_f - K_i = \Delta K$$



Work–Kinetic Energy Theorem

- ▶ The Work–Kinetic Energy Theorem states
$$W_{ext} = K_f - K_i = \Delta K$$
- ▶ When **work is done on** a system **and** the **only change in the system is in its speed**, the net work done on the system equals the change in kinetic energy of the system.
 - W is positive $\Rightarrow$ speed of the system increases
 - W is negative $\Rightarrow$ speed of the system decreases
 - Also valid for changes in rotational speed (later chapters)

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Work–Kinetic Energy Theorem

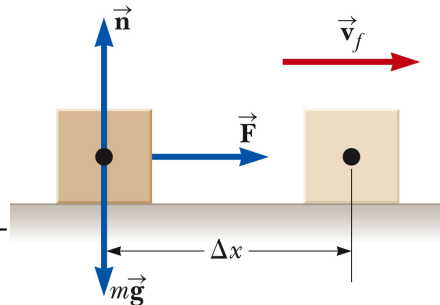
- ▶ The Work–KE theorem **is not valid** when **other changes occur in the system besides speed**, and there are **other interactions** with the environment besides work.
- ▶ The Work–KE theorem relates work to **the speed** of the system, and **NOT to the velocity**.

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Work–Kinetic Energy Theorem – Example

- ▶ The block is the system and three external forces act on it.
- ▶ The normal and gravitational forces do no work since they are perpendicular to the direction of the displacement.



- ▶ $W_{ext} = \Delta K = \frac{1}{2} m v_f^2 - 0$
- ▶ The answer could be checked by modeling the block as a particle and using the kinematic equations.

Example

A force $\vec{F} = (3x^2 \text{ N})\hat{i} + (4 \text{ N})\hat{j}$, with x in meters, acts on a particle changing only the kinetic energy of the particle.

- How much work is done on the particle as it moves from coordinates (2 m, 3m) to (3 m, 0 m)?
- Does the speed of the particle increase, decrease or remain the same?

Example

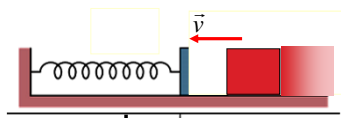
- ▶ A single force acts on a 3.0 kg particle-like object whose position is given by $x = 3.0t - 4.0t^2 + 1.0t^3$, with x in meters and t in seconds. Find the work done on the object by the force from $t = 0$ to $t = 4.0$ s.

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Example

In the figure, a canister of mass $m = 0.40$ kg, slides across a horizontal frictionless surface with a speed $v = 0.50$ m/s. It then runs into and compresses a spring of spring constant $k = 750$ N/m. When the canister is momentarily stopped by the spring, by what distance is the spring compressed?

- $m = 0.40$ kg
- $v_i = 0.50$ m/s
- $v_f = 0$ m/s
- $k = 750$ N/m

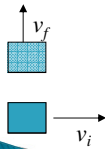


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Example

The only force acting on a 2.0 kg canister that is moving in an xy plane has a magnitude of 5.0 N. The canister initially has a velocity of 4.0 m/s in the positive x direction and some time later has a velocity of 6.0 m/s in the positive y direction. How much work is done on the canister by the 5.0 N force during this time?

- $v_i = 4.0$ m/s
- $v_f = 6.0$ m/s
- $m = 2.0$ kg
- $F = 5.0$ N



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Example

A luge and its rider, with a total mass of 85 kg, emerge from a downhill track onto a horizontal straight track with an initial speed of 37 m/s. If a force slows them to a stop at a constant rate of 2.0 m/s^2

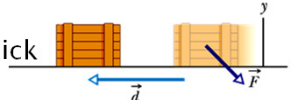
- What is the magnitude F of the net force required?
- What distance d do they travel while slowing down, and
- What work W is done on them (luge and rider) by the force?

- $m = 85$ kg
- $v_i = 37$ /s
- $v_f = 0$ m/s
- $a = 2.0$ m/s²

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Example

During a storm, a crate is sliding across a slick oily parking lot through a displacement $\vec{d} = (-3.0 \text{ m})\hat{i}$ while a steady wind pushes against the crate with a force $\vec{F} = (2.0 \text{ N})\hat{i} + (-6.0 \text{ N})\hat{j}$. The situation and coordinate axes are shown in the Figure.

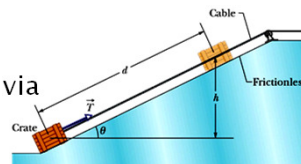


- How much work does this force do on the crate during the displacement?
- If the crate has a kinetic energy of 10 J at the beginning, what is its kinetic energy at the end of the displacement?

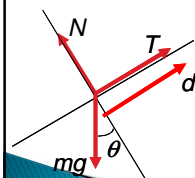
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Problem

A 15.0 kg crate, initially at rest, is pulled, via a cable, a distance $d = 5.70 \text{ m}$ up a frictionless ramp to a height $h = 2.50 \text{ m}$, where it stops.



- How much work is done on the crate by the gravitational force during the lift?
 - How much work is done on the crate by the force T during the lift?
- System?



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Potential Energy

- ▶ Potential energy is energy (!) determined by the configuration of a system in which the components of the system interact by forces.
 - The forces are internal to the system.
 - PE can be associated with only specific types of forces acting between members of a system.
 - E.g. Book–Earth system, spring–block system.

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Gravitational Potential Energy

- ▶ Consider the system of the Earth and the book.
- ▶ Do work on the book by lifting it slowly (no acceleration) through a vertical displacement

$$\Delta \vec{r} = (y_f - y_i) \hat{j}$$

- ▶ The work done on the system must appear as an increase in the energy of the system.

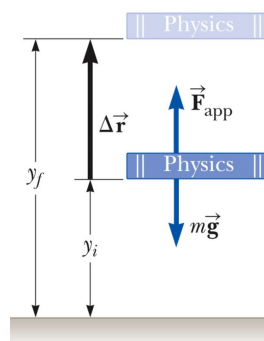
$$W_{\text{ext}} = (\vec{F}_{\text{app}}) \cdot \Delta \vec{r}$$

$$W_{\text{ext}} = (mg \hat{j}) \cdot [(y_f - y_i) \hat{j}]$$

$$= mgy_f - mgy_i$$

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The work done by the agent on the book–Earth system is $mgy_f - mgy_i$.



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Gravitational Potential Energy cont

- ▶ The energy storage mechanism is called *potential energy*.
- ▶ **Gravitational potential energy** is the **energy associated with an object at a given location** above the surface of the Earth.

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Gravitational Potential Energy, final

- ▶ The quantity mgy is identified as the **gravitational potential energy**, U_g .
 - $U_g = mgy$
- ▶ Units are joules (J), and U_g is a scalar
- ▶ g is approx. constant: 9.8 m/s^{-2}
- ▶ Work may change the gravitational potential energy of the system.
 - $W_{\text{ext}} = \Delta U_g$
- ▶ Potential energy is always associated with a system of **two or more** interacting objects.

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Gravitational Potential Energy, Problem Solving

- ▶ The gravitational potential energy depends only on the vertical height of the object above Earth's surface.
- ▶ You must **choose a reference configuration** for which the gravitational potential energy is set equal to some **reference value, normally zero**.
- ▶ **The surface of the Earth is a convenient zero** gravitational potential energy configuration.
- ▶ See Example 7.8 in textbook.

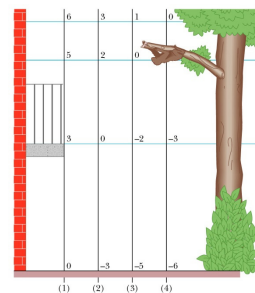
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Example

A 2.0 kg sloth hangs 5.0 m above the ground.

- (a) What is the gravitational potential energy U of the sloth-Earth system if we take the reference point $y = 0$ to be
- At the ground,
 - At the balcony floor (3 m above the ground)
 - At the branch
 - 1.0 m above the limb?
- (b) The sloth drops to the ground. For each of the reference points, what is the ΔU in the potential energy of the Sloth-Earth system.



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Elastic Potential Energy

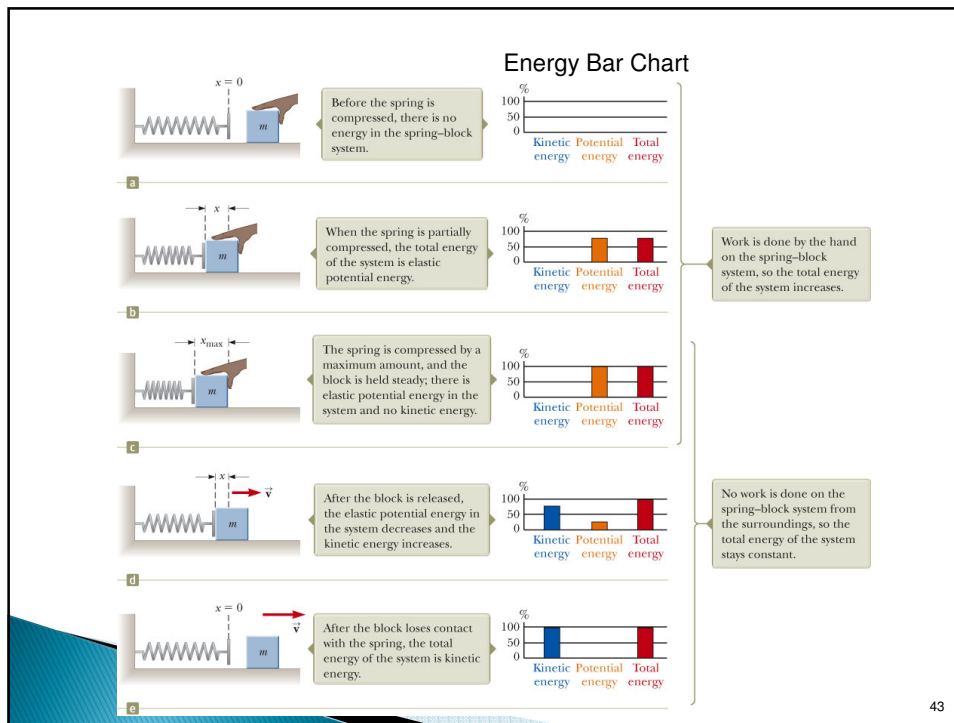
- ▶ **Elastic Potential Energy** is associated with a spring.
- ▶ The force the spring exerts (on a block, for example) is $F_s = -kx$
- ▶ The work done by an external applied force on a spring-block system is
 - $W_{\text{ext}} = \frac{1}{2} kx_f^2 - \frac{1}{2} kx_i^2$
 - The work is equal to the difference between the initial and final values of an expression related to the configuration of the system.
- ▶ The expression for elastic potential energy:
$$U_s = \frac{1}{2} kx^2$$
- ▶ $W_{\text{ext}} = \Delta U_s$

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Elastic Potential Energy cont.

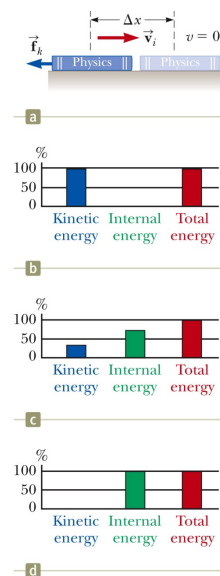
- ▶ The elastic potential energy can be thought of as the energy stored in the deformed spring.
- ▶ The stored PE can be converted into KE.
- ▶ Example Fig 7.16



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Internal Energy

- ▶ The energy associated with an object's temperature is called its *internal energy*, E_{int} .
- ▶ Example: Book slides across the surface.
- ▶ In this example, **the surface is the system**.
- ▶ The **friction** does work and **increases** the internal energy of the surface.
- ▶ **When the book stops**, all of its kinetic energy has been transformed to internal energy.
- ▶ The **total energy remains the same**.



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Conservative Forces

- ▶ Conservative forces have these two properties:
- ▶ The **work done** by a conservative force on a particle moving between any two points is **independent of the path taken by the particle**.
- ▶ The **work done** by a conservative force on a particle **moving through any closed path is zero**.
 - A closed path is one in which the beginning and ending points are the same.
- ▶ Examples of conservative forces:
 - Gravity
 - Spring force

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Conservative Forces, cont

- ▶ We can associate **a potential energy for a system** with any **conservative force acting between members of the system**.
 - In general: $W_{int} = -\Delta U$
 - W_{int} is used as a reminder that the work is done by one member of the system on another member and is internal to the system.
 - **Positive work done by an outside agent** on a system causes an increase in the potential energy of the system.
 - **Work done** on a component of a system **by a conservative force internal** to an isolated system causes a decrease in the potential energy of the system.

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Non-conservative Forces

- ▶ A non-conservative force **does not satisfy the conditions of conservative forces.**
- ▶ Non-conservative forces **acting in a system cause a change in the mechanical energy** of the system.
- ▶ $E_{\text{mech}} = K + U$
 - K includes the kinetic energy of all moving members of the system.
 - U includes all types of potential energy in the system.

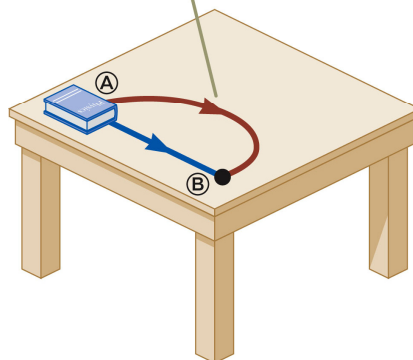
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Non-conservative Forces, cont.

- ▶ Book is displaced from A to B on a surface with friction.
- ▶ The work done against friction is greater along the brown path than along the blue path.
- ▶ Because the work done depends on the path, friction is a non-conservative force.

The work done in moving the book is greater along the brown path than along the blue path.



Conservative Forces and Potential Energy

- ▶ Define a potential energy function, U , such that the work done by a conservative force equals the decrease in the potential energy of the system.
- ▶ The work done by such a force, F , is

$$W_{\text{int}} = \int_{x_i}^{x_f} F_x dx = -\Delta U$$

- ΔU is negative when F and x are in the same direction
- E.g. When an object falls back down to Earth.

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Conservative Forces and Potential Energy

- ▶ The conservative force is related to the potential energy function through.

$$F_x = -\frac{dU}{dx}$$

- ▶ The x component of a conservative force acting on an object within a system equals the negative of the potential energy of the system with respect to x .
 - E.g. 1. block moving up a slope: conservative force is gravity, potential energy is $(-mg\Delta y)$
 - E.g. 2. Block spring system: conservative force is spring force, potential energy is $(\frac{1}{2}kx^2)$
 - This can be extended to three dimensions

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Conservative Forces and Potential Energy – Check

- ▶ Look at the case of a deformed spring:

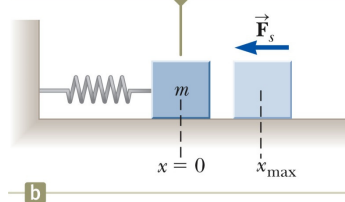
$$F_s = -\frac{dU_s}{dx} = -\frac{d}{dx}\left(\frac{1}{2}kx^2\right) = -kx$$

- This is Hooke's Law and confirms the equation for U
- ▶ U is an important function because a conservative force can be derived from it.

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Energy Diagrams and Equilibrium

The restoring force exerted by the spring always acts toward $x = 0$, the position of stable equilibrium.



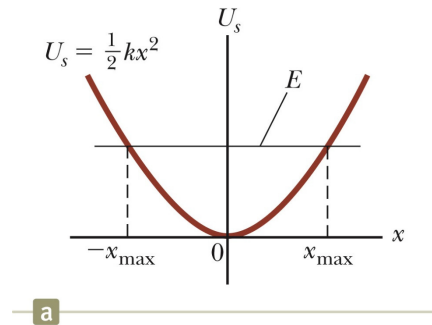
- ▶ Motion in a system can be observed in terms of a **graph of its position and energy**.
- ▶ In a spring-mass system example, the block oscillates between the turning points, $x = \pm x_{\max}$.
- ▶ The block will always accelerate back toward $x = 0$.

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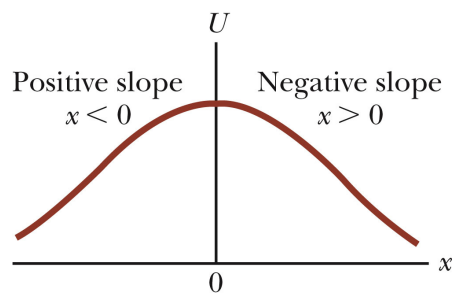
Energy Diagrams and Stable Equilibrium

- ▶ The $x = 0$ position is one of **stable equilibrium**.
 - Any movement away from this position results in a force directed back toward $x = 0$.
- ▶ Configurations of stable equilibrium correspond to those for which $U(x)$ is a **minimum**.
- ▶ $x = x_{\max}$ and $x = -x_{\max}$ are called the **turning points**.



Energy Diagrams and Unstable Equilibrium

- ▶ $F_x = 0$ at $x = 0$, so the particle is in **equilibrium**.
- ▶ For any other value of x , the particle moves away from the equilibrium position.
- ▶ This is an example of **unstable equilibrium**.
- ▶ Configurations of unstable equilibrium correspond to those for which $U(x)$ is a **maximum**.



Neutral Equilibrium

- ▶ **Neutral equilibrium** occurs in a configuration when U is constant over some region.
- ▶ A small displacement from a position in this region will produce neither restoring nor disrupting forces.
- ▶ E.g. ball lying in the middle of a table.

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Potential Energy in Molecules

- ▶ There is potential energy associated with the force between two neutral atoms in a molecule which can be modeled by the **Lennard-Jones** function.

$$U(x) = 4\epsilon \left[\left(\frac{\sigma}{x} \right)^{12} - \left(\frac{\sigma}{x} \right)^6 \right]$$

- ▶ Find the minimum of the function (take the derivative and set it equal to 0) to find the separation for stable equilibrium.
- ▶ The graph of the Lennard-Jones function shows the most likely separation between the atoms in the molecule (at minimum energy).

